Edge Deployment: Plant Health Classification with MobileNet V3 on a Raspberry Pi Zero 2 W

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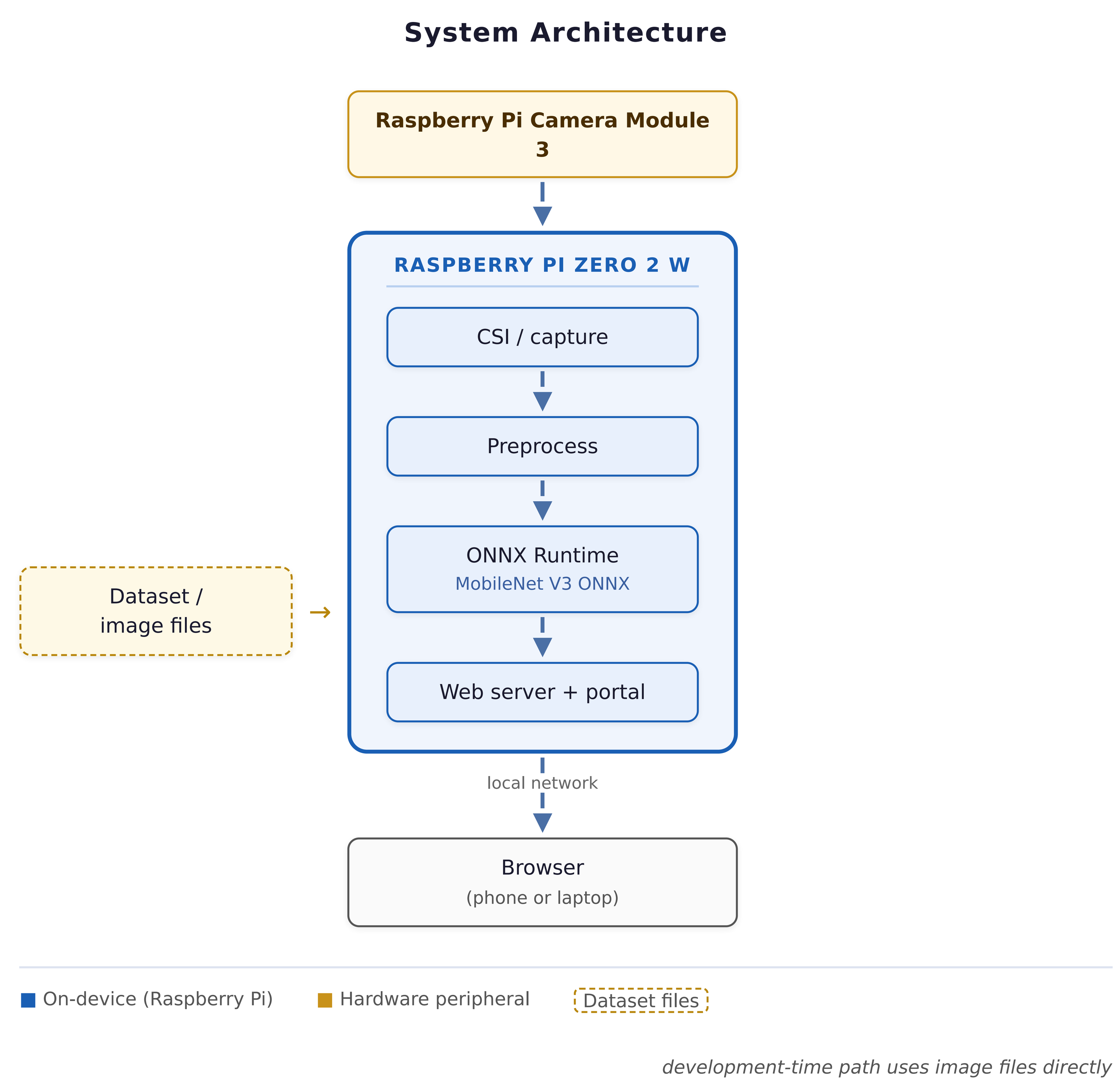
The Problem

Pests and diseases destroy up to **40% of global crops** every year — about **US $220 billion** in losses. Early, on-site detection from a single leaf image protects yield and cuts blanket pesticide use.

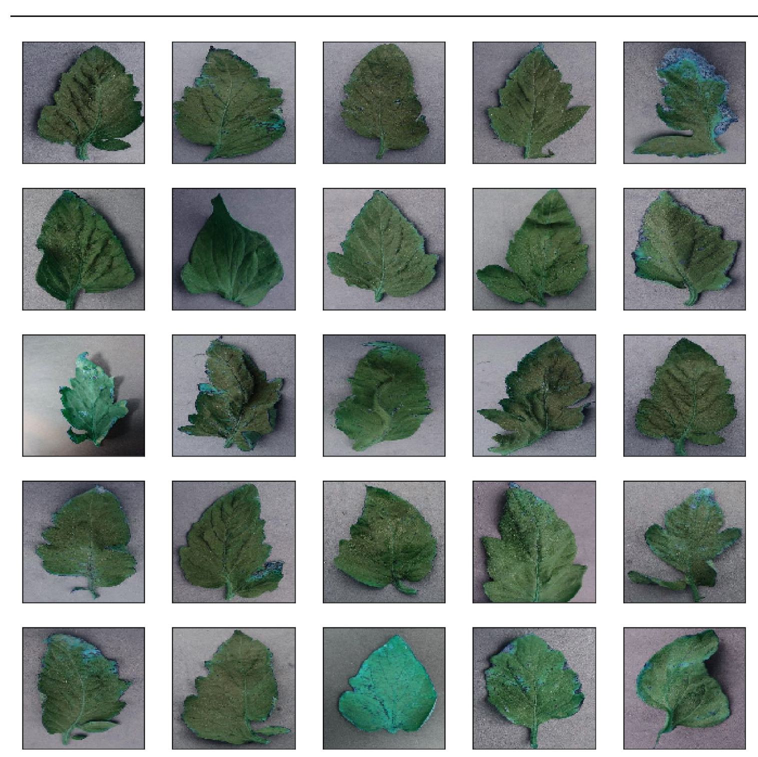
Goal: a self-contained classifier that labels a live camera frame as **Healthy**, **Diseased**, or **Background** — running entirely on a $15, 512 MB Raspberry Pi Zero 2 W with *no* cloud, GPU, or NPU.

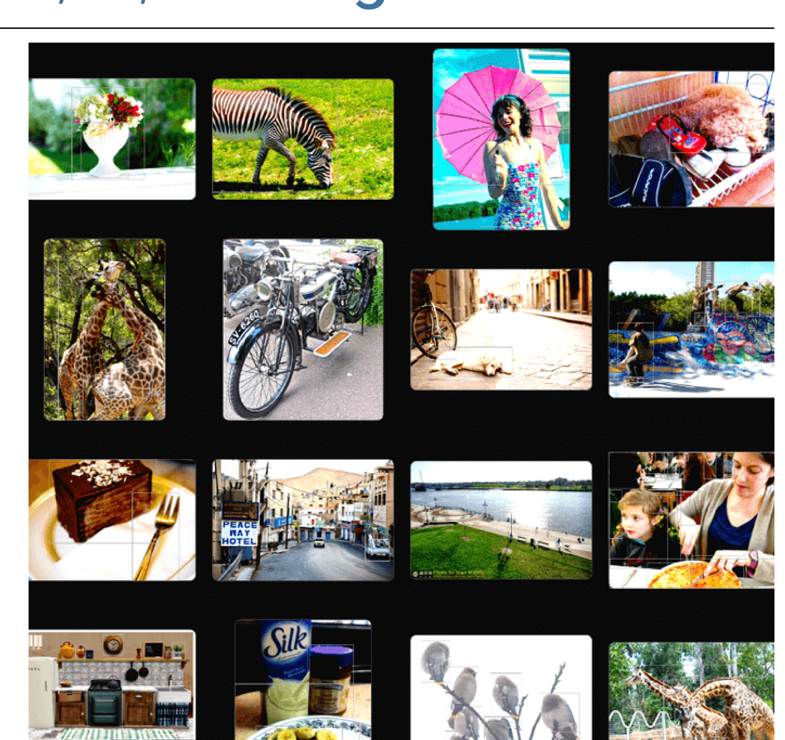
System Architecture



Camera (or stored images) → on-device preprocess → ONNX Runtime → web portal streamed over the LAN.

Dataset — 3 classes, 12,606 images



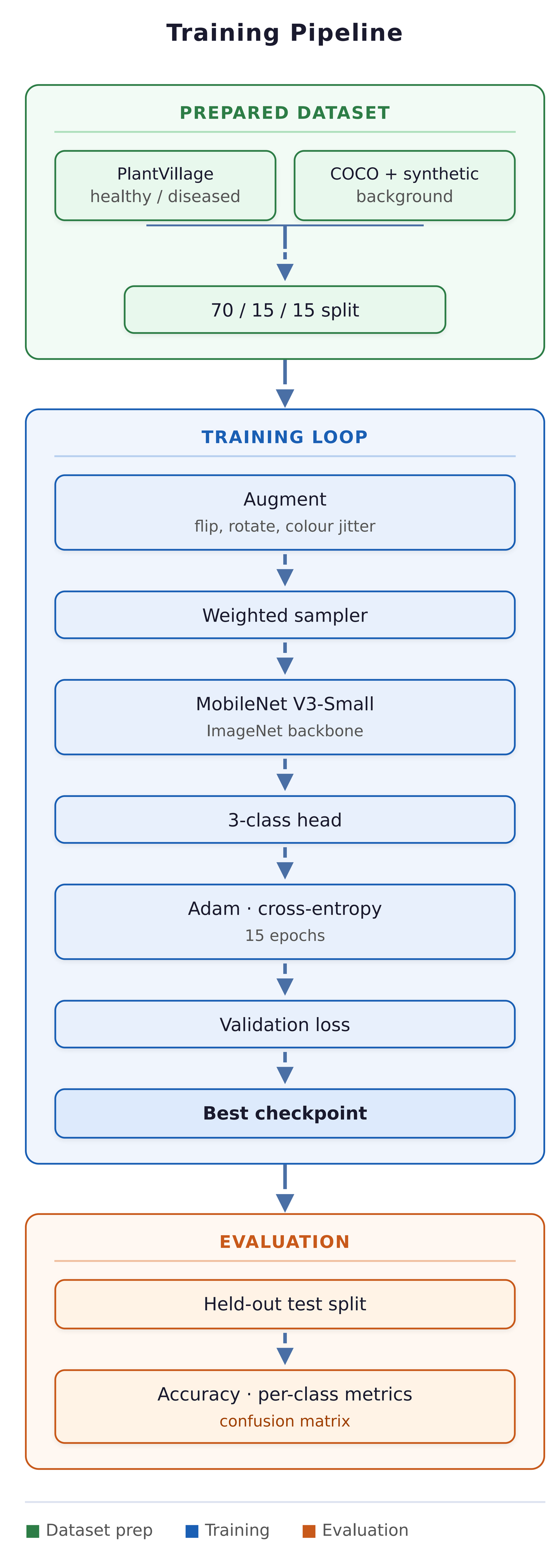


PlantVillage leaves collapsed to healthy / diseased; **COCO 2017** + field negatives form the background class that lets the model reject non-leaf scenes. Split **70 / 15 / 15**.

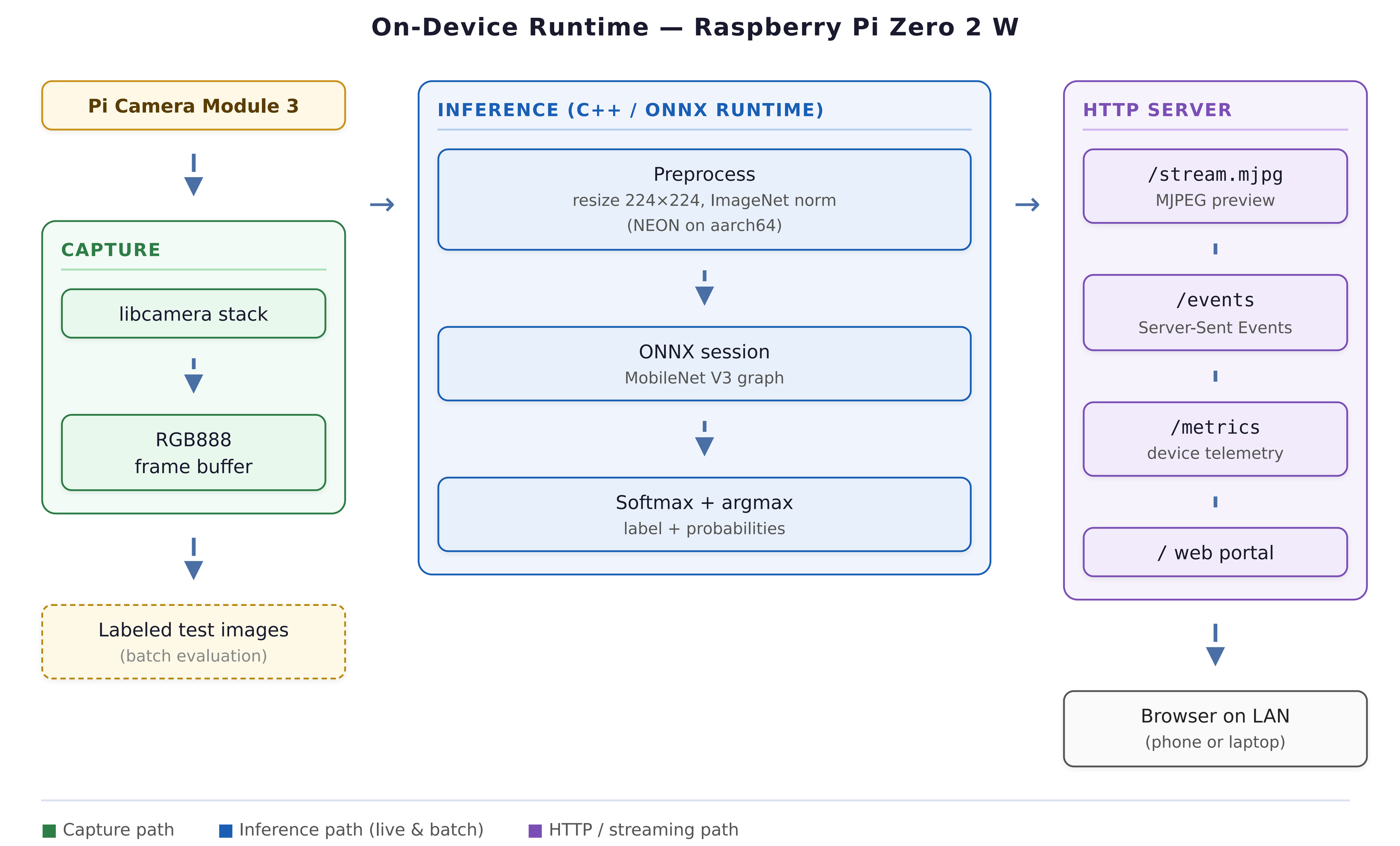
Key Design Choices

- **3-class topology** — a built-in *background* class rejects empty scenes with no extra detection model.
- **FP32 over INT8** — the Cortex-A53 lacks `SDOT`/`UDOT`, so INT8 falls back to slow software; FP32 uses mature NEON FMA instructions.
- **Native C++ runtime** — keeps memory far under the 512 MB ceiling; zero-allocation hot loop.
- **MobileNet V3-Small** — only **1.52 M** parameters.

Training Pipeline



On-Device Runtime (C++)

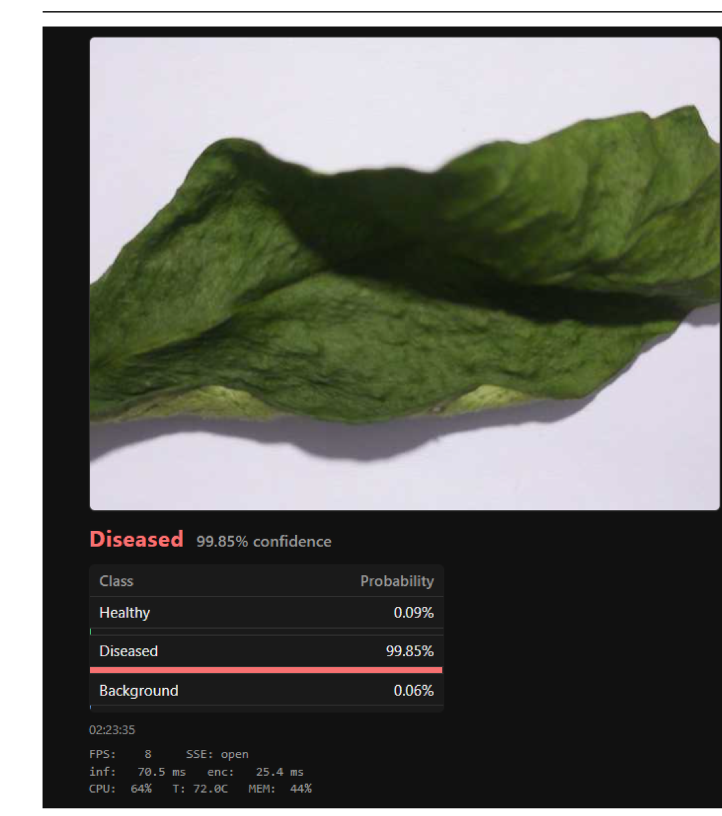


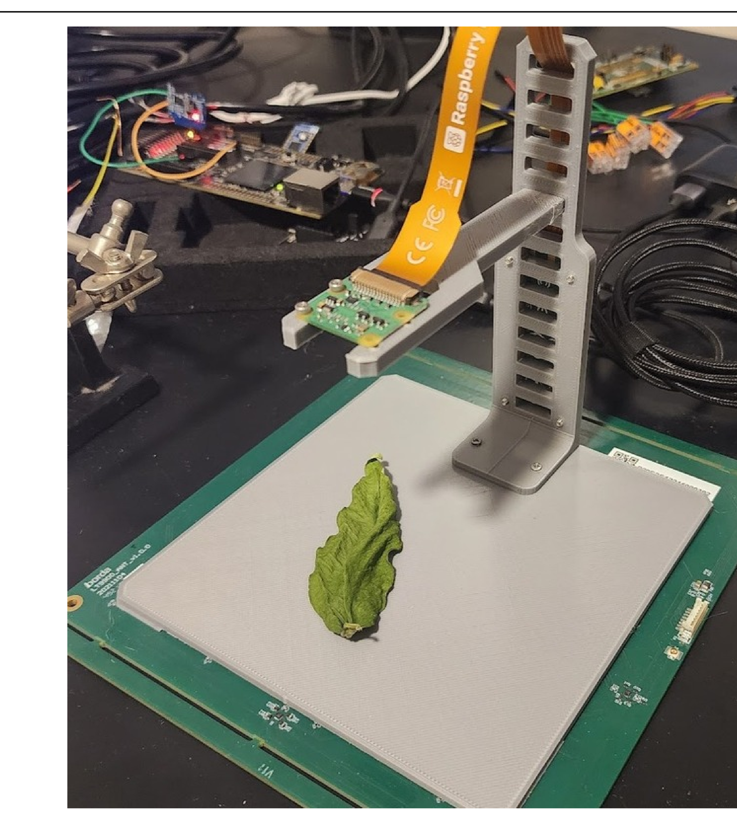
Edge Benchmarks

Build	Split	Acc.	img/s
FP32 (4 thr.)	3-class	**99.81%**	**15.73**
FP32 (1 thr.)	2-class	99.83%	8.03
INT8 (4 thr.)	3-class	97.05%	9.83
INT8 (1 thr.)	3-class	97.21%	6.04

Per-class F1: Healthy 0.995, Diseased 0.998, Background 0.999. INT8 is *slower* here — wrong tool for this silicon.

Live Demo & Test Bench





Web portal shows live prediction, per-class confidence, FPS, and CPU temperature. A 3D-printed PLA bench fixes camera height and lighting for repeatable physical validation.

Conclusion

A native C++ + ONNX Runtime pipeline runs MobileNet V3 at ~**16 img/s** and ~**99.8%** accuracy on a 512 MB Pi Zero 2 W. The background class removes the need for a second model, and FP32 beats INT8 on this CPU. *Future work:* fine-tune on deployment-camera captures and add field validation in live greenhouses.